

BA24/GRAD/XXIV-07/M179D4KY3291

## Certificate of Completion

is proudly presented to

# Muhammad Faris Akbar

9277577 - Universitas Ahmad Dahlan

for successfully completing **Bangkit, specializing in Machine Learning.**

Bangkit is a Google-led academy designed to produce high-caliber technical talent for world-class Indonesian technology companies and startups.

Program Period : February 16, 2024 - June 30, 2024

July 10, 2024

*Dora S.*

**Dora Songco**

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Product Marketing Manager  
Google Indonesia



## STUDENT LEARNING ACHIEVEMENT

Bangkit ID : M179D4KY3291  
Name : Muhammad Faris Akbar  
University : Universitas Ahmad Dahlan

Bangkit Completion : Full Graduate  
Learning Path : Machine Learning  
Capstone Status : Finished

| No | Courses/Specialization/Activities                             | Learning Outcomes                                                                                                                                                                                                                                                                                            | Hours | Score (0-100) | Score Description                                                                                                                                                                                                                                                              |
|----|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Google IT Automation with Python                              | By the end of the course, the student will be able to comprehend how to build Python programs and automate general tasks by writing Python scripts.                                                                                                                                                          | 55    | 85.7          | The student comprehends the knowledge of building Python programs and how to automate general tasks by writing Python scripts.                                                                                                                                                 |
| 2  | Data Analysis with Python                                     | By the end of the course, the students will be able to grasp the key concepts in data analysis and extract insights from data through the analysis process to address business problems effectively.                                                                                                         | 30    | 83.3          | Students are well equipped to grasp the key concepts in data analysis and extract insights from data through the analysis process to address business problems effectively.                                                                                                    |
| 3  | Google Data Analytics                                         | By the end of the course, the student will be able to grasp the learning material related to data cleansing, problem-solving, critical thinking, data ethics, and data visualization.                                                                                                                        | 73    | 88.3          | The student is competent to grasp the learning material related to data cleansing, problem-solving, critical thinking, data ethics, and data visualization.                                                                                                                    |
| 4  | Mathematics for Machine Learning and Data Science             | By the end of the course, the student will have a deep understanding of the math that makes machine learning algorithms work.                                                                                                                                                                                | 58    | 88            | The student exhibits a profound grasp of the mathematical concepts that underlie machine learning algorithms.                                                                                                                                                                  |
| 5  | Machine Learning Specialization                               | By the end of the course, the student will be able to comprehend the fundamental concepts of Machine Learning (Build ML models with NumPy & scikit-learn, supervised models, unsupervised, and use decision trees) and also understand the application of machine learning to solve real-world problems.     | 72    | 89.8          | The student masters the fundamental concepts (Build ML models with NumPy & scikit-learn, supervised models, unsupervised, and use decision trees) and also understands the application of machine learning to solve real-world problems.                                       |
| 6  | Deeplearning.ai TensorFlow Developer Professional Certificate | By the end of the course, the student will be able to apply TensorFlow skills (NLP, Neural Network) to various problems and projects.                                                                                                                                                                        | 79    | 89.8          | The student can apply TensorFlow skills (NLP, Neural Network) to various problems and projects.                                                                                                                                                                                |
| 7  | Tensorflow Data and Deployment                                | By the end of the course, the student will be able to deploy Machine Learning models on the web.                                                                                                                                                                                                             | 47    | 89.8          | The student is adept at deploying Machine Learning models on the web.                                                                                                                                                                                                          |
| 8  | Structuring Machine Learning Projects                         | By the end of the course, the student will be able to execute the end-to-end workflow from the Machine Learning project.                                                                                                                                                                                     | 6     | 81.1          | The student is skilled in executing the end-to-end workflow from the Machine Learning project.                                                                                                                                                                                 |
| 9  | Intro to Generative AI                                        | By the end of the course, the student will be able to grasp key concepts in generative AI.                                                                                                                                                                                                                   | 4     | 89.8          | Students can grasp key concepts in generative AI effectively.                                                                                                                                                                                                                  |
| 10 | TensorFlow: Advance Techniques Specialization                 | By the end of the course, the student will be able to apply advanced techniques in deep learning using TensorFlow.                                                                                                                                                                                           | 67    | 89.8          | Students can effectively apply various advanced techniques in deep learning using TensorFlow.                                                                                                                                                                                  |
| 11 | Preparing for Tensorflow Developer Certification              | By the end of the course, the student will be able to prepare to take the TensorFlow Developer Certificate exam.                                                                                                                                                                                             | 25    | 89.8          | The student has carefully prepared themselves to take the TensorFlow Developer Certificate exam.                                                                                                                                                                               |
| 12 | Capstone / Final Project                                      | By the end of the course, the student will be able to begin stages of a final project, namely developing an application/solution that validates their product development skills and boosts the portfolio.                                                                                                   | 200   | 92.6          | The student is competent to begin stages of a final project, namely developing an application/solution that validates their product development skills and boosts the portfolio.                                                                                               |
| 13 | Soft skill & Career Development                               | By the end of the course, the student will be able to comprehend Life Path, Growth Mindset, The Power of Feedback, Time Management, Critical Thinking, Problem Solving, Adaptability, Resilience, Project Management, Professional Communication, Networking, Digital Branding, and Interview Communication. | 242   | 81.4          | The student thoroughly comprehends Life Path, Growth Mindset, The Power of Feedback, Time Management, Critical Thinking, Problem Solving, Adaptability, Resilience, Project Management, Professional Communication, Networking, Digital Branding, and Interview Communication. |

This is Bangkit-system-generated certificate and valid without signature